

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

8th Semester of B. Pharm. Examination

University Theory Examination November 2016

PH413 Novel Drug Delivery System

Date: 07.11.2016, Monday Time: 1:30 p.m to 4: 30 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any TWO of the following

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| A | (i) Explain osmotic pumps in detail. | 05 |
| | (ii) Explain mechanism of drug release from matrix tablet. | 05 |
| B | Define Liposomes. Classify them based on structural parameters and discuss the applications of the same. | 10 |
| C | Define Iontophoresis. Discuss advantages, disadvantages, side effects and applications of Iontophoresis. | 10 |

Q 2 Attempt any TWO of the following

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|----------|--|-----------|
| A | (i) Discuss solvent evaporation technique for preparation of Microspheres. | 05 |
| | (ii) Explain rotary flash evaporation technique for preparation of Liposomes. | 05 |
| B | Explain various mechanisms of Gastro-retentive drug delivery system. | 10 |
| C | What are hydrogels? Discuss the various approaches to control drug delivery using hydrogels. | 10 |

SECTION- II

Q 3 Attempt any TWO of the following

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|----------|---|-----------|
| A | Explain the four main types of transdermal drug delivery system with diagrammatic representation. | 10 |
| B | Describe the formulation of Fast Dissolving drug delivery system. | 10 |
| C | Differentiate Active targeting of drugs and Passive targeting of drugs. | 10 |

Q 4 Attempt any TWO of the following

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|----------|--|-----------|
| A | Explain diffusion controlled drug delivery system and dissolution controlled drug delivery system. | 10 |
| B | Detailed note on importance of Monoclonal Antibodies in Drug Targeting. | 10 |
| C | Describe in detail Physical, Chemical and Biological approaches of Drug Targeting. | 10 |
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